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Advancing adaptation: How evolving hazards could shape the agenda

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Introduction

Understanding the footprint of climate hazards today and how they will evolve as the global temperature rises is key to decision making, planning, and executing adaptation measures that protect people and economic activity. Thus, the research in our companion report, *Advancing adaptation: Mapping costs from cooling to coastal defenses*, which assesses costs and benefits of adaptation, started with an analysis of the patterns of nine common climate hazards today, and likely changes going forward. We feature that analysis here.

Today, about 40 percent of the Earth's landmass experiences hazards including severe heat, destructive wildfires, prolonged drought, and severe flooding, which are rooted in current local patterns of weather extremes.[1] Heat, wildfires, and drought affect wide swaths of land, while flooding hazards are often more localized, shaped by geography, coastlines, and local rainfall. In addition, some 35 percent of the world's terrain experiences freezing days (see sidebar, "The hazards we examine"). In response, many places around the world have implemented measures to adapt to these hazards.

Yet today's view offers only a snapshot of what is actually a moving picture. Global temperature increases of 1.5°C and 2°C, respectively anticipated by roughly 2030 and 2050, under the current trajectory of global emissions, will cause kaleidoscopic local shifts.[2]

Importantly, change will occur, but not everywhere, not in the same way, and not all at once. While many hazards will spread across more parts of the world, some will not. Freezing is expected to decline, and in some small areas, drought or flood risks may also decrease—even while increasing in others. Hazards may not shift in the same way, either. Many will intensify, lengthen, or happen more frequently as the planet warms, but the scale of the change and where it occurs will vary. For example, regions like Latin America and parts of emerging Asia could face much longer periods of heat stress at 2°C, whereas India, already subject to prolonged heat conditions, is likely to see a smaller relative increase. Finally, exposure to hazards won't increase all at once; more places are expected to become exposed between now and 1.5°C, and still more by 2°C.

The following hazard spotlights explain where, when, and how each type of hazard could change as global temperatures rise from current levels to 1.5°C and 2°C above preindustrial levels. For each hazard, we highlight the change, whether in intensity, frequency, or duration, that matters most for guiding adaptation planning.[3] Naturally, the prevalence of a hazard does not necessarily mean places need to or will adapt to it, but understanding today's patterns of extreme weather and their evolution is the first step in adaptation planning.

In doing this work, we could define hazards in many ways, influencing the results shown here. Since our eventual focus is on understanding adaptation, we define hazards using protection standards typically established in developed economies. And, of course, even though all the hazards examined here are considered meaningful enough to protect against, they are not all created equal, and their impact can be very different. For example, heat stress events are prolonged periods of high heat or humidity, lasting more than a month each year, that broadly affect labor productivity and health. By contrast, heat waves are rare, short-lived, localized extreme heat events that particularly put vulnerable populations at risk.

Since we aren't climate scientists, we rely on external climate models, primarily those used in the sixth assessment report by the Intergovernmental Panel on Climate Change (IPCC).[4] In using these models, we have employed standard techniques to manage uncertainties inherent in climate modeling, though we recognize that important residual uncertainties remain (see the technical appendix for more details). Thus, the results presented here are not high-resolution predictions of climate hazards in specific places around the world. Rather, they are directional assessments of how hazards could evolve, to serve as the basis for our order-of-magnitude assessment of adaptation costs and benefits across the world.

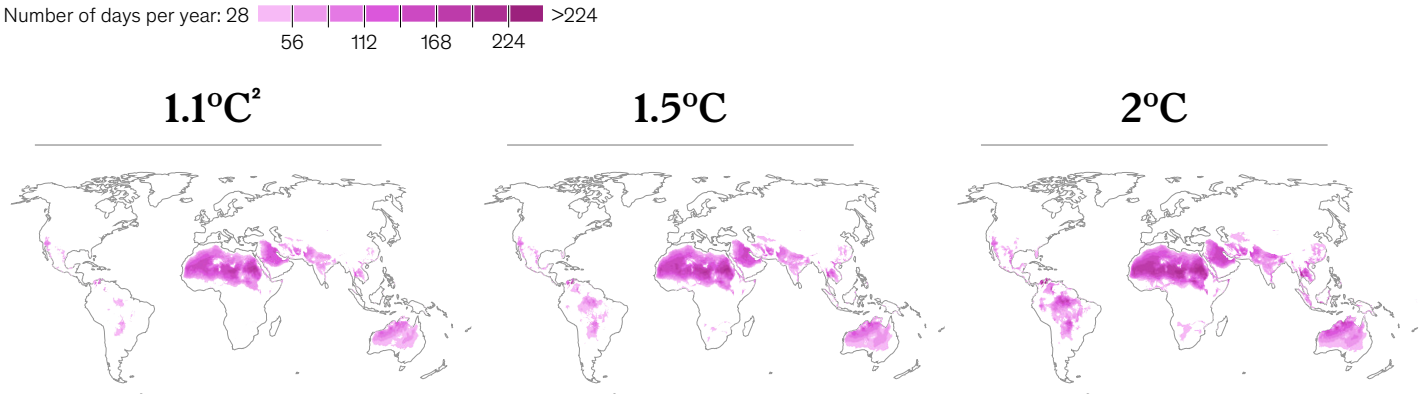
Heat stress: Present across the tropics today, becoming more widespread and prolonged at 2°C

Heat stress refers to sustained periods in an average year of more than four weeks of high temperatures, often combined with humidity, that can significantly reduce labor productivity.[5] Prolonged exposure slows work, increases the need for rest, and raises health risks.[6]

Heat stress exposure is highest today across South Asia, North Africa and the Arabian Peninsula, and parts of Southeast Asia, where some places could have more than 14 weeks a year of heat stress conditions. At 1.5°C of warming relative to preindustrial levels, landmass exposed to heat stress could increase by roughly 25 percent globally compared with today.

At 2°C, global land exposure to heat stress could increase by more than 50 percent compared with today. Areas with little or no exposure now may experience meaningful periods of heat stress. And for regions already exposed today, conditions could intensify as heat stress lasts longer, from roughly 12 weeks today to more than 16 weeks, or an extra month.

Places exposed to heat stress by developed-economy standards, by warming level[1]



Note: This analysis relies on existing and established climate models and techniques, including from the sources below. Hazards are defined based on protection standards established in developed economies. See technical appendix for details.
[1]Heat stress definition: at least 28 days in an average year of either humid heat, when the daily average wet-bulb globe temperature (WBGT, a measure that combines heat and humidity) exceeds 29.4°C, or dry heat, when the daily maximum air temperature exceeds 40°C.
[2]Observations of global temperatures indicate that Earth warmed by 1.1°C relative to preindustrial levels in the decade prior to 2020, which is when our analysis starts.
Source: Coupled Model Intercomparison Project (CMIP6), 2021; NASA NEX-GDDP, 2021; McKinsey Global Institute analysis

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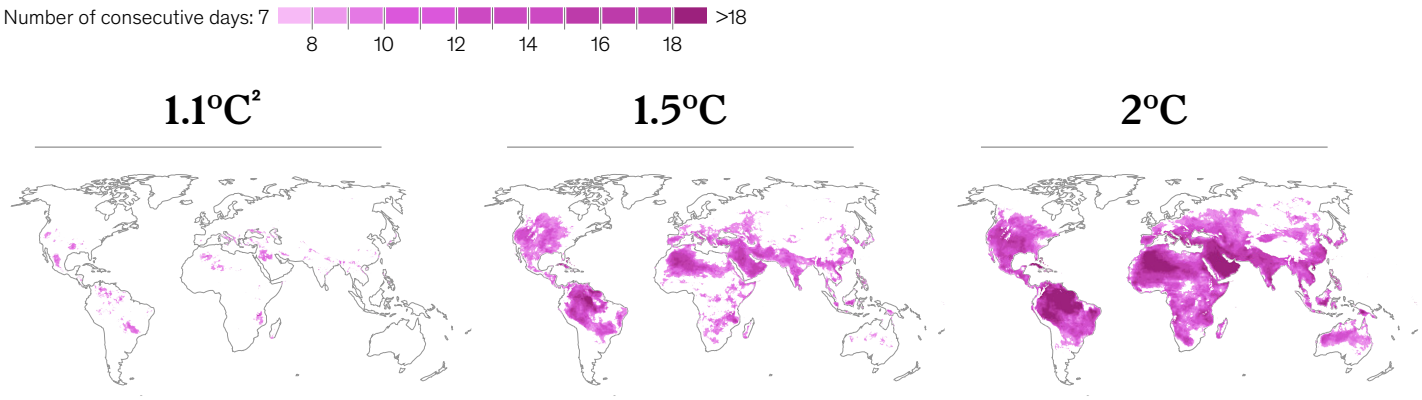
Heat waves: Clustered today in a few places in the tropics and subtropics, spreading at 1.5°C and beyond

Heat can also manifest as heat waves, defined here as rare periods of at least seven consecutive days of unusually high temperatures for a particular place.[7] Not all heat waves across regions are the same. In Nice, France, a week or more of daily highs exceeding 35°C would be considered a heat wave, while in New Delhi, a heat wave event would occur only when temperatures top 45°C. And unlike heat stress, which is an annual occurrence, in any given year neither Nice nor New Delhi might experience such a heat wave. The impact of heat waves can be far lower than heat stress, but when they do occur, they can particularly affect vulnerable populations like elderly people, children, and pregnant women.

Today, parts of South Asia, Southeast Asia, the Arabian Peninsula, and northern parts of South America have a 5 percent likelihood of experiencing such heat waves. In Southeast Asia and along the Arabian Peninsula, such a heat wave can last as long as 12 days. But at 1.5°C of warming, land exposed to a likelihood of heat waves could increase fivefold, and some places could experience heat waves that persist, should they occur, for more than three weeks.

At 2°C, two-thirds of global land could be exposed to a likelihood of such heat waves. Globally, heatwaves of the duration and intensity of today's one-in-20-year event could occur nearly once every four years. However, while large areas may be exposed to such events, protection against heat waves typically relies on relatively low-cost measures, such as early-warning systems and cooling shelters, that can, once put in place, be readily activated when a heat wave occurs.

Places exposed to heat waves by developed-economy standards, by warming level[1]



Note: This analysis relies on existing and established climate models and techniques, including from the sources below. Hazards are defined based on protection standards established in developed economies. See technical appendix for details.
[1]Heat wave definition: a 5% (1-in-20-year) likelihood of at least 7 consecutive days of locally defined high temperatures by today's standards. Locally defined high temperature: the current local 99th percentile daily maximum temperature at 1.1°C, provided it is at or above 32°C.
[2]Observations of global temperatures indicate that Earth warmed by 1.1°C relative to preindustrial levels in the decade prior to 2020, which is when our analysis starts.
Source: Coupled Model Intercomparison Project (CMIP6), 2021; NASA NEX-GDDP, 2021; McKinsey Global Institute analysis

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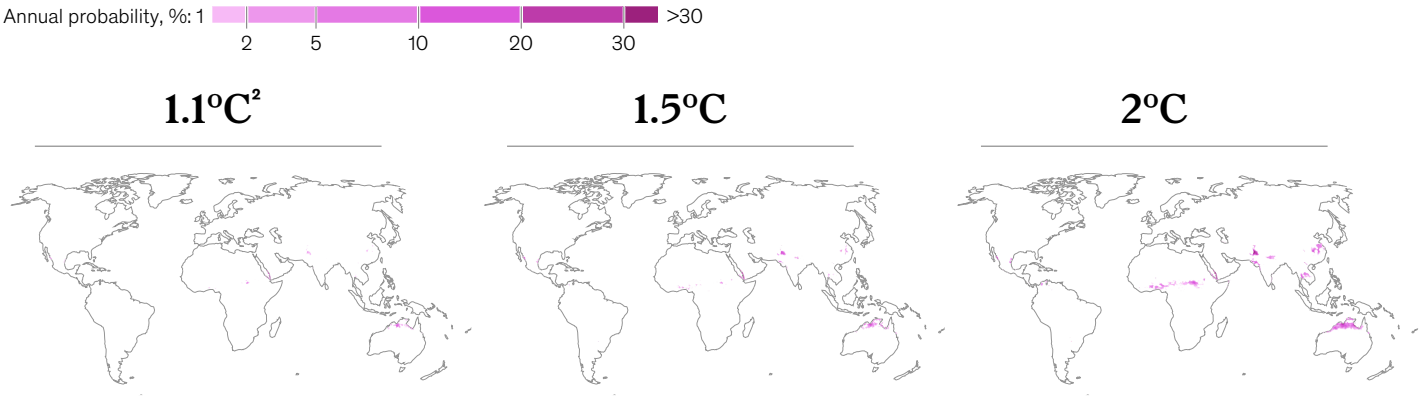
Nonsurvivable heat: Extremely rare today, becoming slightly more widespread and likely at 2°C

Nonsurvivable heat is defined as very rare periods of extremely high, humid heat, such that the body can no longer cool itself by sweating, and during which human survival is at risk without access to cooling.[8] Isolated heat events approaching or exceeding such levels have been recorded in the Persian Gulf and the Indus Valley in recent decades.[9]

Today, only a handful of areas face a very small likelihood of such extreme conditions, including isolated parts of northern Australia, Central and West Africa, South Asia, mainland Southeast Asia, southern China, and around the Red Sea. At 1.5°C of warming relative to preindustrial levels, however, the chance of nonsurvivable heat increases slightly in regions currently at risk.

By 2°C, the total area exposed to this hazard could reach 2 percent of the global landmass. Overall, they would mostly remain very rare events and affect only a very small share of global land. However, in a handful of places—such as a few parts of South Asia and around the Red Sea—the annual probability of an event with such extreme heat could exceed 20 percent then.[10] Measures ranging from air conditioning to early-warning systems could protect against this hazard.

Places exposed to nonsurvivable heat by developed-economy standards, by warming level[1]



Note: This analysis relies on existing and established climate models and techniques, including from the sources below. Hazards are defined based on protection standards established in developed economies. See technical appendix for details.
[1]Nonsurvivable heat definition: a 1% (1-in-100-year) likelihood of at least 24 consecutive hours with minimum wet-bulb temperature (a measure that combines both heat and humidity) at or above 32°C.
[2]Observations of global temperatures indicate that Earth warmed by 1.1°C relative to preindustrial levels in the decade prior to 2020, which is when our analysis starts.
Source: Coupled Model Intercomparison Project (CMIP6), 2021; NASA NEX-GDDP, 2021; McKinsey Global Institute analysis

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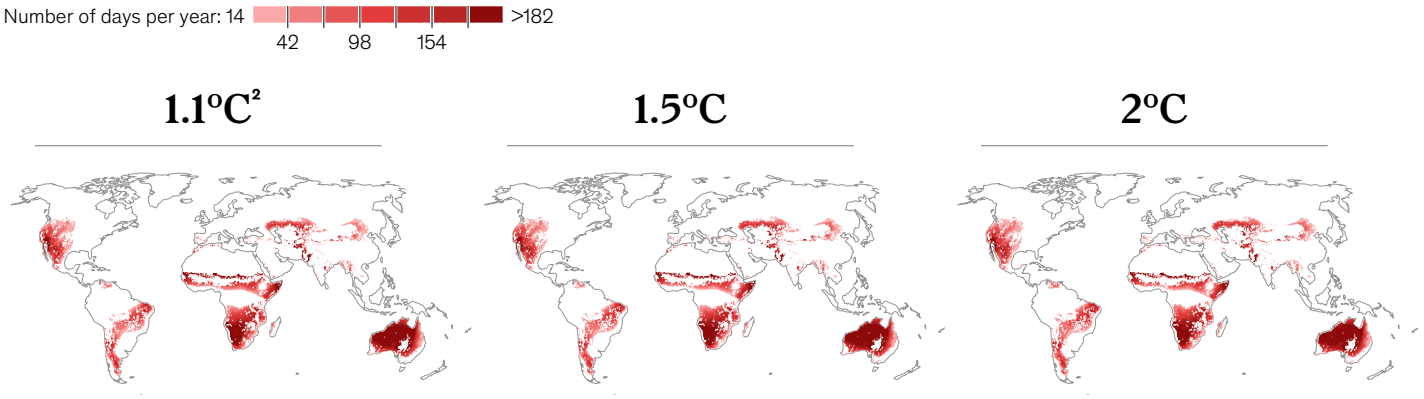
Nonsurvivable heat: Extremely rare today, becoming slightly more widespread and likely at 2°C

Wildfire weather occurs when there are more than two weeks of hot, dry, and windy conditions that heighten the risk of wildfires in places where vegetation can sustain burning.[11] We measure the weather conditions that make wildfires more likely, though of course the occurrence of actual wildfires also requires an ignition event. The effects of these fires can be severe, destroying homes and infrastructure, damaging ecosystems, and releasing smoke that worsens air quality over vast areas. For example, the 2023 Canadian wildfires produced hazardous air quality in cities as far away as New York and Washington, DC.[12]

Wildfire weather is common today across forested and grassland regions worldwide. Parts of southern Africa and Australia could experience more than 180 days a year when conditions are ripe for fire spread.[13] The geographic extent of wildfires increases only modestly at 1.5°C of warming.

At 2°C, land area exposed globally to wildfires may increase by about 10 percent relative to current levels, with affected areas seeing roughly an additional week of such conditions on average.

Places exposed to wildfires by developed-economy standards, by warming level[1]



Note: This analysis relies on existing and established climate models and techniques, including from the sources below. Hazards are defined based on protection standards established in developed economies. See technical appendix for details.
[1]Wildfire-hazard definition: at least 14 days in an average year of hot, dry, and windy conditions—measured as Fire Weather Index (FWI) values exceeding 21 for grassland or 38 for trees—occurring during a single, continuous dry spell (Duff Moisture Code > 20).
[2]Observations of global temperatures indicate that Earth warmed by 1.1°C relative to preindustrial levels in the decade prior to 2020, which is when our analysis starts.
Source: Coupled Model Intercomparison Project (CMIP6), 2021; NASA NEX-GDDP, 2021; McKinsey Global Institute analysis

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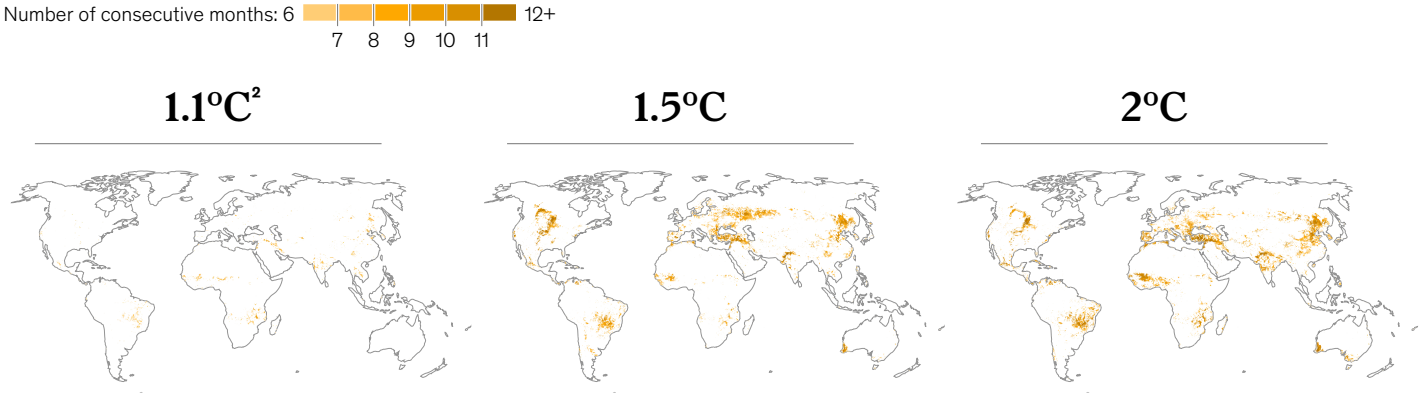
Drought: Present today in some parts of the world, expanding and lengthening at 1.5°C and beyond

Drought, a result of low rainfall and prolonged heat, depletes water in soils, aquifers, and surface waters. We define it as a rare, unusually dry period for a given place, lasting more than six months, that affects agricultural areas.[14] Such events do not occur every year, but when they do, they can disrupt growing cycles, lower yields, and, in severe cases, cause widespread harvest failure. For example, the 2012 drought in the United States caused an estimated $30 billion in agricultural losses as corn and soybean harvests were reduced.[15]

Today, about 13 percent of global agricultural land has a 5 percent chance of having droughts of this kind, primarily in Brazil, the Middle East, the Sahel and southern Africa, Greater China, India, and smaller pockets in the western United States and Mexico. Exposed areas can experience six to eight dry months in a rare drought. The areas exposed to such a rare drought could increase by almost three times today's exposure at 1.5°C of warming.

At 2°C, drought could increase by almost four times today's exposure and in some regions, a rare drought may last more than 12 dry months. A drought of the duration and intensity of today's one-in-20-year event could, on average, become almost a one-in-10-year event in the future. About 4 percent of all areas that experience a change in drought exposure may be no longer exposed as rainfall patterns shift.

Places exposed to drought by developed-economy standards, by warming level[1]



Note: This analysis relies on existing and established climate models and techniques, including from the sources below. Hazards are defined based on protection standards established in developed economies. See technical appendix for details.
[1]Drought definition: a 5% (1-in-20-year) likelihood of at least 6 consecutive months with dry conditions by today's standards. Dry conditions definition: a value below −1.3 in the 1.1°C climatology on the Terrestrial Water Storage Drought Standardization Index (TWS-DSI, a measure that characterizes variations in total water available at a local scale, combining all potential sources and sinks of freshwater, including atmospheric water, soil moisture, groundwater, and surface water).
[2]Observations of global temperatures indicate that Earth warmed by 1.1°C relative to preindustrial levels in the decade prior to 2020, which is when our analysis starts.
Source: Inter-Sectoral Impact Model Intercomparison Project Phase 2b (ISIMIP2b), 2017; McKinsey Global Institute analysis

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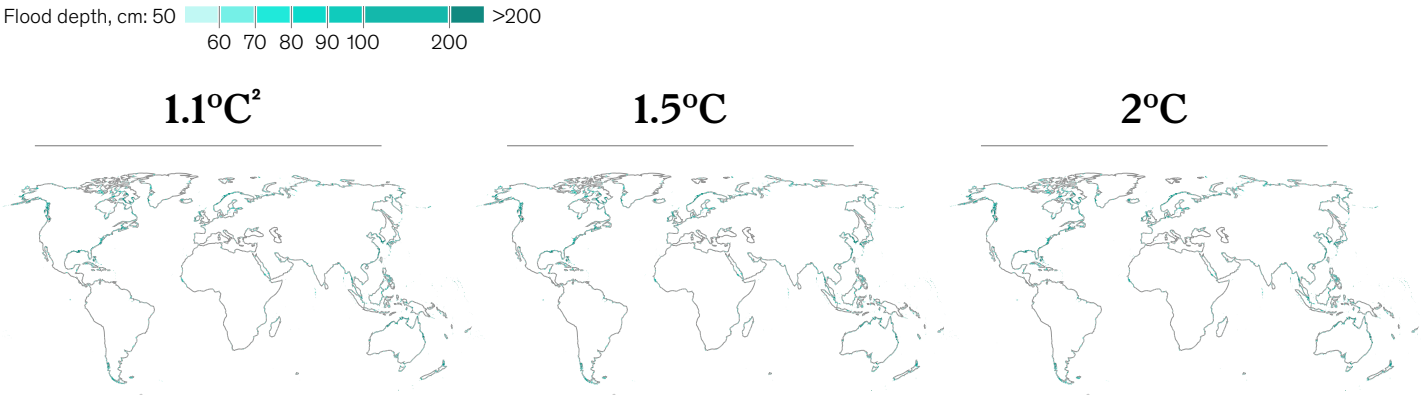
Coastal flooding: Rare events today expected to occur more often across many low-lying regions at 2°C, with some intensity increase

Coastal flooding—defined here as a very rare event causing at least knee-deep inundation—occurs when high tides and storm surges force seawater to cover normally dry coastal land.[16] In 2022, Hurricane Ian pushed seawater inland across southwest Florida, causing dozens of fatalities and extensive damage to homes and infrastructure far from the shoreline.[17]

Today, such flooding can affect low-lying coastal areas worldwide, including many small island nations, parts of northern Europe, South and Southeast Asia, and the coast of the Gulf of Mexico, which the United States calls the Gulf of America. Some more places would be exposed to such rare events as sea levels further rise with warming.[18] In these places, even modest surges or high tides can trigger inundation. At 1.5°C of warming, coastal land exposed to these very rare floods could expand by about 6 percent, and today's flood event may occur five times more often on average.

By 2°C at 2050, exposure could increase by roughly 15 percent compared with today. Today's one-in-100-year event could become a one-in-13-year event by that time. Average depths of the one-in-100-year event could become 10 percent higher on average, though with wide regional variation.

Places exposed to coastal flooding by developed-economy standards, by warming level[1]



Note: This analysis relies on existing and established climate models and techniques, including from the sources below. Hazards are defined based on protection standards established in developed economies. See technical appendix for details.
[1]Coastal flooding definition: a 1% (1-in-100-year) likelihood of a flood event with inundation of at least 50 cm. Coastal flooding exposure considers only the transient sea-level response that corresponds to warming levels reached at specific points in time: 1.1°C in 2020, 1.5°C in 2030, and 2.0°C in 2050.
[2]Observations of global temperatures indicate that Earth warmed by 1.1°C relative to preindustrial levels in the decade prior to 2020, which is when our analysis starts.
Source: Fathom Global Flood Map Fathom 3.0, 2021; McKinsey Global Institute analysis

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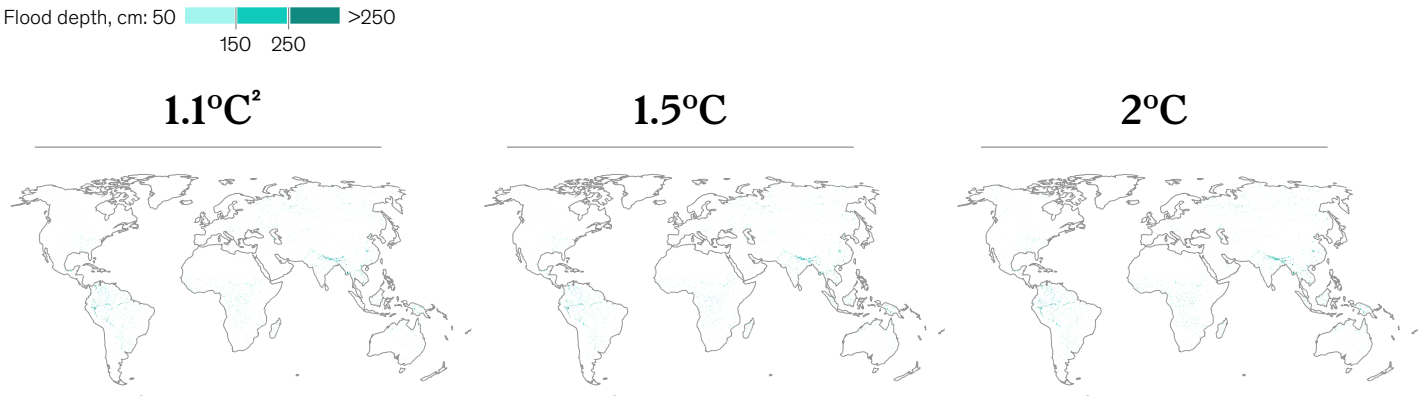
Riverine flooding: Present today, with slightly greater extent and some frequency and intensity increase at 2°C

Riverine flooding, defined here as a very rare event bringing at least knee-deep inundation, occurs when rivers and streams overflow onto normally dry land due to excessive rainfall or snowmelt.[19] In July 2021, extreme flooding along the Meuse and Rhine rivers in Europe killed more than 220 people and caused about €46 billion in losses, primarily in infrastructure.[20]

Today, places exposed to riverine flooding are widespread across many of the world's major river basins, with flood depths in such rare events often exceeding one meter. The share of land exposed to riverine flooding could increase by about 2 percent at 1.5°C of warming .

At 2°C, the global land exposed could increase by about 4 percent. Extreme riverine floods could become slightly more frequent then, with today's one-in-100-year event occurring on average once in 80 years. The average flood depth in exposed regions may rise modestly, by about five to ten centimeters, though with wide regional variation. Isolated areas in temperate and polar basins may experience reduced flood risk as rainfall and snowpack patterns shift.

Places exposed to riverine flooding by developed-economy standards, by warming level[1]



Note: This analysis relies on existing and established climate models and techniques, including from the sources below. Hazards are defined based on protection standards established in developed economies. See technical appendix for details.
[1]Riverine flooding definition: a 1% (1-in-100-year) likelihood of a flood event with inundation of at least 50 cm.
[2]Observations of global temperatures indicate that Earth warmed by 1.1°C relative to preindustrial levels in the decade prior to 2020, which is when our analysis starts.
Source: Fathom Global Flood Map Fathom 3.0, 2021; McKinsey Global Institute analysis

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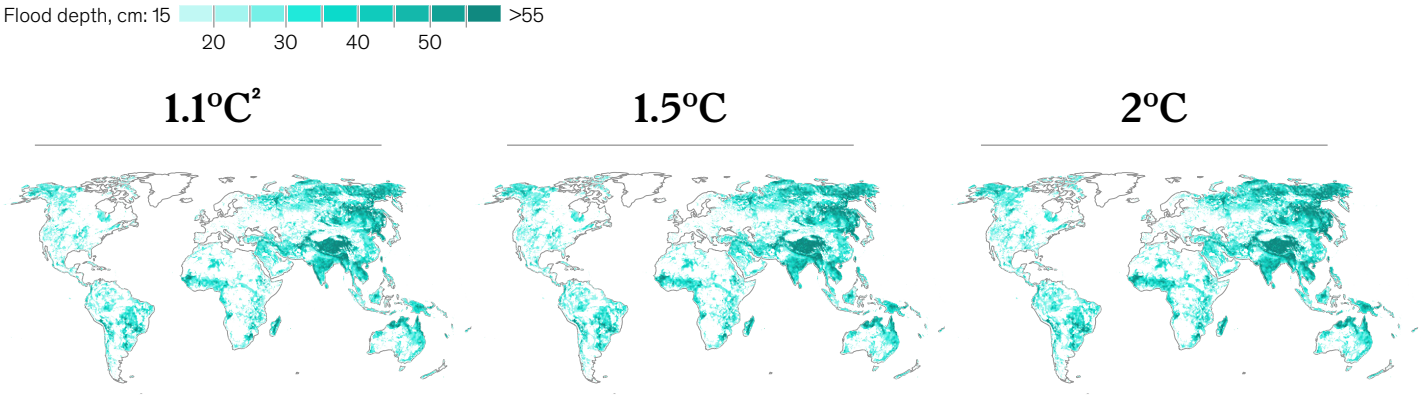
Excessive rainfall flooding: Widespread today, expanding in reach at 2°C

Excessive rainfall (pluvial) flooding, defined here as a rare event causing at least ankle-deep inundation, occurs when typically short bursts of heavy rainfall overwhelm the capacity of soil or drainage systems to absorb or channel water, causing water to pool on normally dry land or pavement.[21] In 2023, a record downpour in Hong Kong flooded subway stations and halted stock exchange trading, with damages exceeding $100 million.[22]

Rainwater flooding is already widespread globally. Many locations experience rare events where flood depths are 15 centimeters or more, enough to inundate streets, damage crops and property, and disrupt urban systems. At 1.5°C, exposed landmass that has a 5 percent chance of flooding from excessive rainfall could expand by nearly 10 percent compared with today, though isolated areas may see decreases.

At 2°C, global land exposure could increase by more than 20 percent. At this level of warming, a one-in-20-year event could become slightly more frequent, roughly one in 17 years, and the global average depth of such a flood would rise only slightly, by two centimeters.

Places exposed to excess-rainfall flooding by developed-economy standards, by warming level[1]



Note: This analysis relies on existing and established climate models and techniques, including from the source below. Hazards are defined based on protection standards established in developed economies. See technical appendix for details.
[1]Excess-rainfall flooding definition: a 5% (1-in-20-year) likelihood of a flood event with inundation of at least 15 cm.
[2]Observations of global temperatures indicate that Earth warmed by 1.1°C relative to preindustrial levels in the decade prior to 2020, which is when our analysis starts.
Source: Fathom Global Flood Map Fathom 3.0, 2021; McKinsey Global Institute analysis

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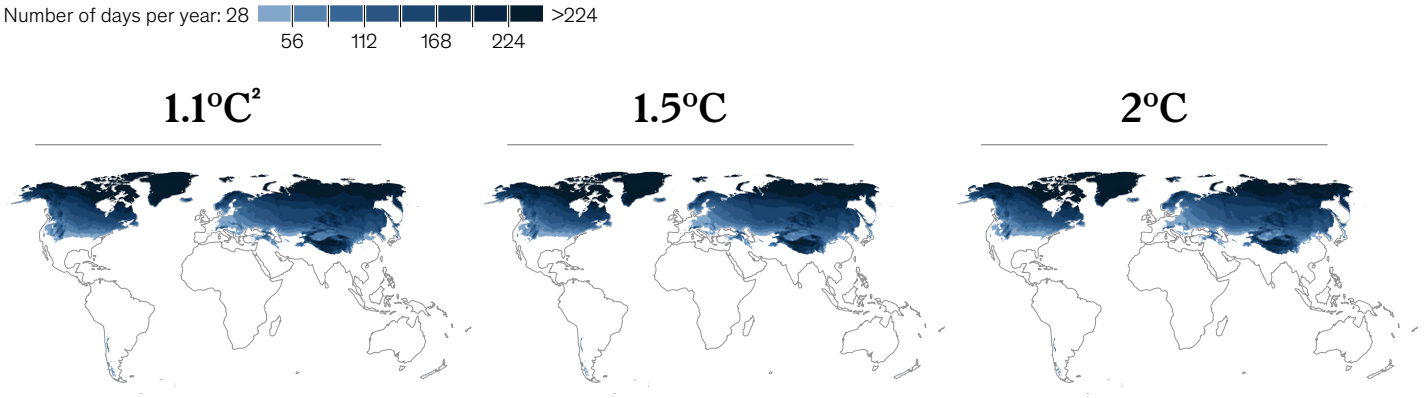
Freezing days: Common today across temperate and polar regions, becoming less widespread with warming at 1.5°C and beyond

As global temperatures rise, the freezing days hazard—at least four weeks of days when the average temperature is below 0°C—could occur in fewer places.[23] This may bring benefits as it could reduce cold-related deaths and crop damage. However, it also brings risks such as shrinking glacial freshwater reserves and thawing permafrost.[24]

Today, freezing days are common across northern continental interiors, including most of Canada, the northern United States, and northern Europe into Russia. At 1.5°C of warming, about 2 percent of land currently experiencing freezing days is expected not to have them.

At 2°C, the global land exposed could decrease by about 3 percent. Areas with decreasing exposure to freezing days then will also be more densely populated. The biggest decreases may occur in parts of North America, with smaller declines in Central Europe and Northeast Asia. Places that continue to see freezing conditions as the global temperature rises are expected to face them for shorter durations each year.

Places exposed to freezing by developed-economy standards, by warming level[1]



Note: This analysis relies on existing and established climate models and techniques, including from the sources below. Hazards are defined based on protection standards established in developed economies. See technical appendix for details.
[1]Freezing days definition: at least 28 days in an average year when the daily average temperature is below 0°C.
[2]Observations of global temperatures indicate that Earth warmed by 1.1°C relative to preindustrial levels in the decade prior to 2020, which is when our analysis starts.
Source: Coupled Model Intercomparison Project (CMIP6), 2021; NASA NEX-GDDP, 2021; McKinsey Global Institute analysis

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Explore the full report

Advancing adaptation: Mapping costs from cooling to coastal defenses estimates the costs of climate adaptation using a granular, geospatial analysis and examining 20 proven, cost-effective adaptation measures.

Acknowledgments

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This research contributes to our mission to help business and policy leaders understand the forces transforming the global economy. As with all MGI research, it is independent and has not been commissioned or sponsored in any way by any business, government, or other institution. While we benefited greatly from the variety of perspectives we gathered, we have independently formed and articulated our views in this report. Any errors are our own.

Endnotes

1 Observations of global temperatures indicate that Earth warmed by 1.1°C relative to preindustrial levels in the decade prior to 2020, which is when our analysis starts and what we refer to as "current" or "today's" climate conditions. (The average of the most recent estimates for the 2015–24 decade is 1.24°C of warming relative to the global temperature between 1850 and 1900.) In this report, warming levels refer to long-run average temperatures rather than temperatures in any given year, in line with typical climate analysis. See "Summary for policymakers," in *Climate Change 2021: The Physical Science Basis*, Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, Cambridge University Press, 2021; and "Indicators of global climate change 2024: Annual update of key indicators of the state of the climate system and human influence," in *Earth System Science Data*, Copernicus Publications, 2025. Our analysis excludes Antarctica because it is not populated.

2 We assessed 25 current-trajectory emissions scenarios, based on policies announced or implemented, from the IPCC, NGFS, and IEA to reflect the wide range of climate estimates. These scenarios suggest that 1.5°C of warming relative to preindustrial levels would be reached somewhere between 2025 and 2035, when warming is measured as a 20-year average centered on a given year and relative to a preindustrial average. In 2030, temperatures across the 25 scenarios range from 1.45°C to 1.55°C. In the IPCC scenarios, the median level of warming by 2050 is 2.0°C, and in the more recent NGFS and IEA scenarios, it is 2°C. By 2100, the temperature estimates from the same set of models are 3.0°C (average across IPCC scenarios, with a maximum of 3.7°C), 2.9°C (NGFS and IEA Current Policies), and 2.5°C (IEA Stated Policies). The IPCC and NGFS scenarios are based on the assumption that currently announced policies are preserved. For NGFS, this covers national climate policies that were legislated and supported by instruments as of March 2024. For IPCC scenarios, the specific policies reflected vary by model. The IEA Current Policies scenario (2025), by contrast, considers only policies already adopted in legislation and regulation, assuming no future changes, even where governments have indicated their intention to do so, and takes a cautious perspective on the pace at which new energy technologies are deployed and integrated into the energy system. The IEA Stated Policies scenario draws on a wider interpretation of the policy environment, incorporating not only enacted measures but also formally proposed ones and other official strategy documents that signal the intended policy direction. For details, see IEA, *World Energy Outlook 2025*, November 2025. On these trajectories, warming is projected to rise beyond 2°C in the second half of this century. Because some adaptation measures take more than a decade to implement, anticipating needs at least for 2050 and a 2°C world now is prudent, even as the world works to reduce emissions. Moreover, some measures implemented today—for example, stormwater networks—can have multiple-decade lifetimes, making it prudent to consider how hazards could evolve a few decades out, to avoid costly retrofits down the road.

3 These hazards occur along a continuum of intensity, frequency, or duration, requiring a defined threshold for understanding the hazard. Thresholds can be defined in various ways. Since our focus is on understanding and quantifying adaptation costs, we draw on protection standards commonly established in developed economies. Of course, the hazards we examine may not manifest in the same way or have comparable impacts. For instance, coastal flood protection standards in developed economies typically aim to withstand a one-in-100-year flood event, while pluvial flood protection standards often correspond to a one-in-20-year flood event. We further validated these thresholds using literature from the IPCC on typical hazard definitions and their impacts. For more details on our approach to understanding climate hazards and their behavior with warming, refer to the technical appendix. The findings presented apply only to the specific hazard metrics and definitions used in this research. The view of how hazards evolve over time and adaptation needs could look different for other thresholds and hazard definitions.

4 *Climate Change 2023: Synthesis Report*, Contribution of Working Groups I, II, and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, Intergovernmental Panel on Climate Change, 2023.

5 We define a place as being exposed to heat stress if it experiences at least 28 days in an average year of either humid heat, when the daily average wet-bulb globe temperature (WBGT, a measure that combines heat and humidity) exceeds 29.4°C (corresponding to roughly 25 percent of labor productivity loss), or dry heat, when the daily maximum air temperature exceeds 40.0°C. See Peter Brode et al., "Estimated work ability in warm outdoor environments depends on the chosen heat stress assessment metric," *International Journal of Biometeorology*, April 2017; and Brenda Jacklitsch et al., *Occupational exposure to heat and hot environments*, US Department of Health and Human Services, February 2016.

6 Wojciech Szewczyk, Ignazio Mongelli, and Juan-Carlos Ciscar, "Heat stress, labour productivity and adaptation in Europe—a regional and occupational analysis," *Environmental Research Letters*, October 2021, Volume 16, Number 10; Kerstin K. Zander et al., "Heat stress causes substantial labour productivity loss in Australia," *Nature Climate Change*, July 2015, Volume 5; Anton Orlov et al., "Economic costs of heat-induced reductions in worker productivity due to global warming," *Global Environmental Change*, July 2020, Volume 63; and Tord Kjellstrom, Ingvar Holmer, and Bruno Lemke, "Workplace heat stress, health and productivity—an increasing challenge for low and middle-income countries during climate change," *Global Health Action*, 2009, Volume 2, Number 1.

7 We define a place as being exposed to heat waves if it experiences at least seven consecutive days of locally defined high temperatures by today's standards, with a 5 percent (1-in-20-year) likelihood. Locally defined high temperature is defined as the current local 99th percentile daily maximum temperature in that place, provided it is at or greater than 32°C.

8 We define a place as being exposed to nonsurvivable heat if it experiences at least 24 consecutive hours when the minimum WBGT is at or above 32°C with at least a 1 percent (1-in-100-year) likelihood.

9 Colin Raymond, Tom Matthews, and Cascade Tuholske, "Evening humid-heat maxima near the southern Persian/Arabian Gulf," *Communications Earth & Environment*, October 2024; and Jenix Justine et al., "The diurnal variation of wet-bulb temperatures and exceedance of physiological thresholds relevant to human health in South Asia," *Communications Earth & Environment*, July 2023.

10 While metrics and thresholds of nonsurvivable heat vary, these locations and the likely magnitude of change compared with today are broadly consistent with other studies. See Jeremy S. Pal and Elfatih A. B. Eltahir, "Future temperature in southwest Asia projected to exceed a threshold for human adaptability," *Nature Climate Change*, October 2015; Eun-Soon Im, Jeremy S. Pal, and Elfatih A. B. Eltahir, "Deadly heat waves projected in the densely populated agricultural regions of South Asia," *Science Advances*, August 2017; and Fahad Saeed, Carl-Friedrich Schleussner, and Moetasim Ashfaq, "Deadly heat stress to become commonplace across South Asia already at 1.5°C of global warming," *Geophysical Research Letters*, March 2021.

11 We define a place as being exposed to wildfire weather if an average year sees at least 14 days of dangerously hot, dry, and windy conditions measured as Fire Weather Index (FWI) values exceeding 21 for grassland or 38 for trees, occurring during a single, continuously dry spell (Duff Moisture Code > 20).

12 "Notice of Code Purple, very unhealthy air for D.C. region," Metropolitan Washington Council of Governments, June 8, 2023; and "Transcript: Mayor Adams briefing on NYC's air quality," New York City Office of the Mayor, June 7, 2023.

13 Direct numerical comparisons with the literature are limited by differing metrics and thresholds, but published sources qualitatively describe long, dry seasons greater than six months in parts of southern Africa and Australia. See Justin Perry et al., *Savanna fire regimes and biodiversity impacts in northern Australia*, Commonwealth Scientific and Industrial Research Organisation, June 2021; and Brian John Huntley, "The arid savanna biome," in *Ecology of Angola: Terrestrial Biomes and Ecoregions*, Springer, 2023.

14 There is no single agreed definition of drought in the scientific community. Some studies focus on short dry spells, others on multiyear events, and there is large variance in how often droughts are assumed to occur in the literature. We define a place as being exposed to drought if it experiences at least six consecutive months of dry conditions by today's standards with a 5 percent (1-in-20-year) likelihood. Dry conditions are measured locally and defined as when the value of the Terrestrial Water Storage Drought Standardization Index (TWS-DSI) metric, a measure that characterizes local variations in total water available, combining all potential sources and sinks of freshwater including atmospheric water, soil moisture, groundwater and surface water, falls below –1.3.

15 The US National Oceanic and Atmospheric Administration (NOAA) describes 2012 as the most extensive US drought since the 1930s, with moderate to extreme drought occurring across more than half the country. Affected states spanned the West and the central farm belt. "Monthly climate reports: Drought: Annual 2012," NOAA National Centers for Environmental Information, 2013; Bradley R. Rippey, "The U.S. drought of 2012," *Weather and Climate Extremes*, December 2015, Volume 10, Part A; and US Department of Agriculture, National Agricultural Statistics Service, "Crop production down in 2012 due to drought," January 11, 2013.

16 We define a place as being exposed to coastal flooding if inundation of at least 50 centimeters occurs with a 1 percent (1-in-100-year) likelihood.

17 Lisa Bucci et al., "Hurricane Ian," National Hurricane Center Tropical Cyclone Report, April 2023.

18 Coastal flood exposure reflects only the "transient" sea-level response that corresponds to warming levels reached at specific points in time, 1.1°C at 2020, 1.5°C at 2030, and 2.0°C at 2050. It does not account for any additional "committed" rise that occurs over much longer timescales as oceans expand and ice sheets gradually melt, even if global temperature increases stabilize.

19 We define a place as being exposed to riverine flooding if inundation of at least 50 centimeters occurs with a 1 percent (1-in-100-year) likelihood.

20 Frank Kreienkamp et al., "Rapid attribution of heavy rainfall events leading to the severe flooding in Western Europe during July 2021," World Weather Attribution, July 2021; "Munich Re expects rising reinsurance prices in Europe," Munich Re, October 18, 2021; and "Hope following the catastrophe," German Federal Foreign Office, December 12, 2021.

21 We define a place as being exposed to rainwater flooding if inundation of at least 15 centimeters occurs with a 5 percent (1-in-20-year) likelihood.

22 "Hong Kong's historic storm could cost city at least $136 million," *Straits Times*, November 14, 2024.

23 We define a place as being exposed to freezing days hazard if it experiences at least one month in an average year when the average temperature is below 0°C.

24 *Climate Change 2021: The Physical Science Basis*, Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, Cambridge University Press, August 2021; Intergovernmental Panel on Climate Change (IPCC), *IPCC Special Report on the Ocean and Cryosphere in a Changing Climate*, Cambridge University Press, 2019.

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